

מטלה 1

חלק ב:

Decision Tree:

1. `DecisionTreeClassifier()`
Accuracy Score: 0.92
Precision Score:0.9696969696969697
Recall Score:0.9142857142857143
F1 Score:0.9411764705882353
Confusion Matrix:
[[14 1]
 [3 32]]
2. `DecisionTreeClassifier(criterion='entropy')`
Accuracy Score: 0.94
Precision Score:0.9444444444444444
Recall Score:0.9714285714285714
F1 Score:0.9577464788732394
Confusion Matrix:
[[13 2]
 [1 34]]
3. `DecisionTreeClassifier(random_state=0)`
Accuracy Score: 0.92
Precision Score:0.9428571428571428
Recall Score:0.9428571428571428
F1 Score:0.9428571428571428
Confusion Matrix:
[[13 2]
 [2 33]]
4. `DecisionTreeClassifier(max_depth=5)`
Accuracy Score: 0.92
Precision Score:0.9428571428571428
Recall Score:0.9428571428571428
F1 Score:0.9428571428571428
Confusion Matrix:
[[13 2]
 [2 33]]
5. `DecisionTreeClassifier(criterion='entropy', max_depth=5)`
Accuracy Score: 0.94
Precision Score:0.9444444444444444
Recall Score:0.9714285714285714
F1 Score:0.9577464788732394
Confusion Matrix:
[[13 2]
 [1 34]]

6. `DecisionTreeClassifier(criterion='entropy', max_depth=5, random_state=0)`
Accuracy Score: 0.92
Precision Score:0.9428571428571428
Recall Score:0.9428571428571428
F1 Score:0.9428571428571428
Confusion Matrix:
[[13 2]
 [2 33]]
7. `DecisionTreeClassifier(criterion='entropy', splitter='random')`
Accuracy Score: 0.96
Precision Score:1.0
Recall Score:0.9428571428571428
F1 Score:0.9705882352941176
Confusion Matrix:
[0 15]
 [2 33]]
8. `DecisionTreeClassifier(max_depth=5, splitter='random')`
Accuracy Score: 0.98
Precision Score:1.0
Recall Score:0.9714285714285714
F1 Score:0.9855072463768116
Confusion Matrix:
[0 15]
 [1 34]]

Random Forest:

1. **RandomForestClassifier()**
Accuracy Score: 0.96
Precision Score:0.9714285714285714
Recall Score:0.9714285714285714
F1 Score:0.9714285714285714
Confusion Matrix:
[[14 1]
 [1 34]]
2. **RandomForestClassifier(criterion='entropy')**
Accuracy Score: 0.96
Precision Score:0.9714285714285714
Recall Score:0.9714285714285714
F1 Score:0.9714285714285714
Confusion Matrix:
[[14 1]
 [1 34]]
3. **RandomForestClassifier(random_state=0)**
Accuracy Score: 0.96
Precision Score:0.9714285714285714
Recall Score:0.9714285714285714
F1 Score:0.9714285714285714
Confusion Matrix:
[[14 1]
 [1 34]]
4. **RandomForestClassifier(max_depth=5)**
Accuracy Score: 0.96
Precision Score:0.9714285714285714
Recall Score:0.9714285714285714
F1 Score:0.9714285714285714
Confusion Matrix:
[[14 1]
 [1 34]]
5. **RandomForestClassifier(criterion='entropy', max_depth=5)**
Accuracy Score: 0.96
Precision Score:0.9714285714285714
Recall Score:0.9714285714285714
F1 Score:0.9714285714285714
Confusion Matrix:
[[14 1]
 [1 34]]
6. **RandomForestClassifier(criterion='entropy', max_depth=5, random_state=0)**
Accuracy Score: 0.96
Precision Score:0.9714285714285714

Recall Score:0.9714285714285714

F1 Score:0.9714285714285714

Confusion Matrix:

```
[[14  1]
```

```
 [ 1 34]]
```

7. RandomForestClassifier(bootstrap=False)

Accuracy Score: 0.96

Precision Score:0.9714285714285714

Recall Score:0.9714285714285714

F1 Score:0.9714285714285714

Confusion Matrix:

```
[[14  1]
```

```
 [ 1 34]]
```

8. RandomForestClassifier(criterion='entropy',bootstrap=False)

Accuracy Score: 0.96

Precision Score:0.9714285714285714

Recall Score:0.9714285714285714

F1 Score:0.9714285714285714

Confusion Matrix:

```
[[14  1]
```

```
 [ 1 34]]
```

9. RandomForestClassifier(criterion='entropy',warm_start=True)

Accuracy Score: 0.96

Precision Score:0.9714285714285714

Recall Score:0.9714285714285714

F1 Score:0.9714285714285714

Confusion Matrix:

```
[[14  1]
```

```
 [ 1 34]]
```

AdaBoost:

1. AdaBoostClassifier()

Accuracy Score: 0.96

Precision Score:0.9714285714285714

Recall Score:0.9714285714285714

F1 Score:0.9714285714285714

Confusion Matrix:

```
[[14  1]
 [ 1 34]]
```

2. AdaBoostClassifier(n_estimators=100)

Accuracy Score: 0.98

Precision Score:0.9722222222222222

Recall Score:1.0

F1 Score:0.9859154929577465

Confusion Matrix:

```
[[14  1]
 [ 0 35]]
```

3. AdaBoostClassifier(random_state=0)

Accuracy Score: 0.96

Precision Score:0.9714285714285714

Recall Score:0.9714285714285714

F1 Score:0.9714285714285714

Confusion Matrix:

```
[[14  1]
 [ 1 34]]
```

4. AdaBoostClassifier(learning_rate=5.0)

Accuracy Score: 0.92

Precision Score:0.918918918918919

Recall Score:0.9714285714285714

F1 Score:0.9444444444444444

Confusion Matrix:

```
[[12  3]
 [ 1 34]]
```

5. AdaBoostClassifier(n_estimators=100, learning_rate=5.0)

Accuracy Score: 0.84

Precision Score:0.8292682926829268

Recall Score:0.9714285714285714

F1 Score:0.8947368421052632

Confusion Matrix:

```
[[ 8  7]
 [ 1 34]]
```

6. AdaBoostClassifier(n_estimators=75, random_state=0)

Accuracy Score: 0.98

Precision Score:0.9722222222222222

Recall Score:1.0

F1 Score:0.9859154929577465

Confusion Matrix:

```
[[14  1]
```

```
 [ 0 35]]
```

7. **AdaBoostClassifier(n_estimators=50, random_state=0)**

Accuracy Score: 0.96

Precision Score:0.9714285714285714

Recall Score:0.9714285714285714

F1 Score:0.9714285714285714

Confusion Matrix:

```
[[14  1]
```

```
 [ 1 34]]
```